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METHODS AND APPARATUS FOR DYNAMIC MUSIC CREATION IN VIDEO CONTENT CREATION APPLICATIONS

Abstract

Systems, apparatus, articles of manufacture, and methods are disclosed for dynamic music creation in video content creation applications. An example apparatus to generate audio tracks for a video content creation application disclosed herein includes interface circuitry, machine-readable instructions, and at least one processor circuit to be programmed by the machine-readable instructions to: analyze a video to determine video characteristics; generate a prompt for a machine learning model based on the video characteristics and at least one user preference; provide the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and provide the audio track to a video content creation application.

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Background/Summary

RELATED APPLICATION [0001] This application claims the benefit of priority to Patent Cooperation Treaty (PCT) Application No. PCT/CN2024/093324 filed May 15, 2024. Patent Cooperation Treaty (PCT) Application No. PCT/CN2024/093324 is incorporated by reference in its entirety.

BACKGROUND
[0002] When editing video content, it is common to add a background music track to the video content. When used appropriately, background music can enhance the video content viewing experience.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] FIG. **1** is a block diagram of an example environment in which an example audio creation circuitry operates to generate audio tracks for media content.
[0004] FIG. **2** is a block diagram of an example implementation of the audio creation circuitry of FIG. **1**.
[0005] FIG. **3** illustrates example voice audio isolation processing performed by the audio creation circuitry of FIG. **1**.
[0006] FIG. **4** is an example cold-lighting video analyzed by the audio creation circuitry of FIG. **1**.
[0007] FIG. **5** is an example warm-lighting video analyzed by the audio creation circuitry of FIG. **1**.
[0008] FIG. **6** illustrates person and/or scene detection as performed by the audio creation circuitry of FIG. **1**.
[0009] FIG. **7** is a flowchart representative of example machine-readable instructions and/or example operations that may be executed, instantiated, and/or performed by example programmable circuitry to implement the audio creation circuitry of FIG. **2**.
[0010] FIG. **8** is a table illustrating an example text management data structure generated by the audio creation circuitry of FIG. **1**.
[0011] FIG. **9** is a block diagram illustrating an example workload schedule implemented by the audio creation circuitry of FIG. **1**.
[0012] FIG. **10** is an example user interface implemented by the audio creation circuitry of FIG. **1**.
[0013] FIG. **11** is a block diagram of an example processing platform including programmable circuitry structured to execute, instantiate, and/or perform the example machine-readable instructions and/or perform the example operations of FIG. **7** to implement the audio creation circuitry of FIG. **2**.
[0014] FIG. **12** is a block diagram of an example implementation of the programmable circuitry of FIG. **11**.
[0015] FIG. **13** is a block diagram of another example implementation of the programmable circuitry of FIG. **11**.
[0016] FIG. **14** is a block diagram of an example software/firmware/instructions distribution

platform (e.g., one or more servers) to distribute software, instructions, and/or firmware (e.g., corresponding to the example machine-readable instructions of FIG. **7**) to client devices associated with end users and/or consumers (e.g., for license, sale, and/or use), retailers (e.g., for sale, re-sale, license, and/or sub-license), and/or original equipment manufacturers (OEMs) (e.g., for inclusion in products to be distributed to, for example, retailers and/or to other end users such as direct buy customers).

[0017] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale.

DETAILED DESCRIPTION

[0018] Adding a background audio track to video content can enhance the viewing experience for viewers, especially when the background music aligns with the mood or theme of the video. However, today the process for a content creator is difficult. The content creator has to pick through thousands of third-party created songs and music to try to find something suitable, and of course, most of the music available is copyrighted by the original musician who created it. Video content publishing platforms (e.g., YouTube, Vimeo, etc.) allow content creators to use music from most well-known artists in the videos uploaded by the content creators, however, the publishing platforms typically de-monetize the videos that use copyrighted music for the content creator and all monetization of the video content goes entirely to the musician, not the content creator. Thus, by simply using a single copyrighted song, the content creator loses 100% of the monetization value of their content. Therefore, content creators desire to use copyright-free and/or generated music. Currently, content creators must manually generate new music, generate music via a text-to-music method, or pay a third party to generate music.

[0019] Methods and apparatus disclosed herein analyze video content and automatically generate copyright-free audio. Such methods and apparatus accept user preferences as inputs from content creators to tailor audio track generation to the preferences of the content creator. In some examples, methods and apparatus analyze the video content to determine video content characteristics (e.g., source media attributes) to use in audio track generation. Example embodiments generate prompts for machine learning models based on the video characteristics and the user preferences. In some examples, methods and apparatus disclosed herein provide the audio track to media creation applications (e.g., Adobe Premiere Pro, Davinci Resolve, AVID, Vegas Magix, PowerDirector, etc.) used by the content creator. In examples disclosed herein, a user is enabled to update user preferences for generation of an updated audio track based on the updated user preferences.

[0020] Examples disclosed herein identify a variety of video characteristics, including, but not limited to: whether, and where, the video contains dialogue; color tone of the video; a type of activity being performed in the video; speed of action in the video; environment(s) detected in the video; person(s) detected in the video; number of people or activity detection; happiness detection (e.g., using facial detection to determine if a scene is a happy scene or a somber scene); and type of video (e.g., landscape, cartoon, or feature film).

[0021] Methods and apparatus disclosed herein generate audio tracks for the video content based in part on user preferences including, but not limited to: duration (e.g., scene music segmentation); background music priority; music preference (e.g., genre preference); and weights for the video characteristics. Methods and apparatus disclosed herein cause presentation of a user interface to allow a user to input user preferences for audio generation.

[0022] FIG. **1** is a block diagram of an example environment **100** in which an example audio creation circuitry **120** operates to generate audio tracks for media creation applications. The example environment includes an example content creator **105**, an example source media, and example media creation platform, the example audio creation circuitry **120**, an example created media **125**, an example published location **130**, and example viewers **140**.

[0023] The example media creation platform **115** is a content creation application. Content creation applications are digital tools or services that enable users to create, edit, and publish media content. Example content creation applications may include, for example, Adobe Premiere Pro, Davinci Resolve, AVID, Vegas Magix, PowerDirector, etc. In some examples, the media content creation platform **115** is an application hosted on a device of the content creator **105**. In other examples, the media creation platform **115** is a web application, a cloud application, etc. The example media creation platform receives the source media **110** and audio tracks generated by the audio creation circuitry **120** to generate the created media **125**, which can ultimately be published at the published location **130**.

[0024] The example audio creation circuitry **120** is an application. In some examples, the audio creation circuitry **120** is a computing device such as a server, a personal computer, a workstation, a mobile device, or any other type of computing device. In some examples, the audio creation circuitry **120** is an Application-Specific Integrated Circuit, an XPU, a Field-Programmable Gate Array, and other any other processor circuitry and/or combination of processor circuitries. In the illustrated example of FIG. **1**, the audio creation circuitry **120** is a component of the media creation platform **115**. In other examples, the audio creation circuitry **120** is a standalone application.

[0025] The example audio creation circuitry **120** analyzes the source media **110** and automatically generates copyright-free audio tailored to the content of the source media **110** and the preferences of the content creator **105**. The example audio creation circuitry **120** analyzes the source media **110** to determine video content characteristics to use in audio track generation. In some examples, the audio creation circuitry **120** executes a machine learning model to analyze the source media **110**. The example audio creation circuitry **120** identifies a variety of video characteristics, including, but not limited to: whether, and where, the clip contains dialogue; color tone of the video; a type of activity being performed in the video; speed of action in the video; environment(s) detected in the video; person(s) detected in the video; number of people or activity detection; happiness detection (e.g., using facial detection to determine if a scene is a happy scene or a somber scene); and type of video (e.g., landscape, cartoon, or feature film). The example audio creation circuitry **120** adjusts variables used to generate audio tracks based on the identified video characteristics. For example, the audio creation circuitry **120** can perform voice isolation processing of the source media **110** to identify portions of the source media **110** that include dialogue. In those examples, the audio creation circuitry **120** uses that information to generate an audio track that plays quieter/less distracting music to ensure the dialogue is clear and audible even with the background music.

[0026] In some examples, the audio creation circuitry **120** performs color analysis on the source media **110** to determine the color grading/warmth of the video. In those examples, the audio creation circuitry **120** uses the color tone of the source media **110** to derive a mood of the video and generate an audio track that fits the mood. In some examples, the audio creation circuitry **120** performs people detection using facial recognition and/or or other object recognition techniques. In those examples, the audio creation circuitry **120** can generate audio to assign to specific people, such that particular instruments or types of music are used with greater emphasis for particular subjects. In some examples, the audio creation circuitry **120** identifies the type of activity being performed. For example, the audio creation circuitry **120** can identify that a sport is being played and generates an audio track with a genre of music that matches that sport. Environment detection can include, for example, identifying the scene takes place by a waterfall or an ocean, which can be used by the audio creation circuitry **120** to generate an audio track that is suitable for flowing water.

[0027] The example audio creation circuitry **120** generates audio tracks for the video content based in part on user preferences including, but not limited to: duration (e.g., scene music segmentation); background music priority; music preference (e.g., genre preference); a preferred music generating machine learning model; weights and/or other instructions for use of the video characteristics. For example, the content creator **105** can indicate how the music should be segmented across the source media **110**. For example, the source media **110** may have an average clip length of six seconds and the average scene may be from 20-120 seconds—meaning that each scene includes 3-20 independent camera clips. It might be desirable that the music for a particular scene be consistent, however when there are only 20 seconds per scene, switching to new music every scene can be undesirable and continuation of the same music from one scene to another is preferred. The audio creation circuitry **120** enables the content creator **105** to control which scenes are grouped together, and which are separated if the content creator **105** wishes to do so. In some examples, duration preferences include a minimum and/or maximum track length. In some examples, the audio creation circuitry **120** can override user preferences based on the identified video characteristics.

[0028] In some examples, the audio creation circuitry **120** causes presentation of a user interface to allow a content creator **105** to input user preferences for audio generation. In some examples, the audio creation circuitry **120** allows the content creator **105** to adjust parameters on a sliding scale. In some examples, the audio creation circuitry **120** performs natural language comprehension to extract meaning from a text input from the content creator **105**. For example, the content creator **105** can input text stating a desired music genre be used for a specific person appearing in the source media **110** and the audio creation circuitry **120** converts the input to variables to be used in generation of the audio track. In some examples, the audio creation circuitry **120** allows a user to update user preferences for generation of an updated audio track based on the updated user preferences.

[0029] The example audio creation circuitry **120** generates prompts for machine learning models based on the video characteristics and the user preferences. The example audio creation circuitry **120** creates prompts that include a plurality of entries, some of the entries related to the user preferences and some of the entries related to the identified video characteristics. In some examples, each entry has three elements: 1) an index to indicate whether the entry is related to the user input or an identified video characteristic; 2) an indication of the specific variable of user input or video characteristic (e.g., music duration or type); and 3) a variable value from user input or video characteristic (e.g., 20 s for duration, or pop for music type, “happy” was detected from scene). In some examples, audio creation circuitry **120** executes a programming language (e.g., Python, c++ or other programming language) to extract the text to form one or two sentences as the prompt of AI music generator. In an example where the source media **110** includes a video of a little girl smiling and the user inputs a preference of “10 s” duration and “pop” music genre, the audio creation circuitry **120** generates the example prompt: “10 s duration, pop happy music.” The complexity of the prompt is mostly decided by the video and the complexity of the user preferences input by the content creator **105**. Furthermore, this structure can be also easily extended based on the needs.

[0030] In some examples, the audio creation circuitry **120** executes a machine learning model to generate an audio track based on the prompt generated from the video characteristics and user preferences. In some examples, the audio creation circuitry **120** sends the prompt to an external music generating machine learning model. In some examples, the audio creation circuitry **120** provides the audio track to media creation platform **115**. In some examples, the audio creation circuitry **120** provides the audio track to the content creator **105**.

[0031] The example published location **130** is video content publishing platform such as YouTube, Vimeo, etc. The example published location **130** allows the content creator **105** to publish the created media **125** for viewing by viewers **140**.

[0032] FIG. **2** is a block diagram of an example implementation of the audio creation circuitry **120** of FIG. **1** to generate audio tracks for media content. The audio creation circuitry **120** of FIG. **2** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by programmable circuitry such as a Central Processor Unit (CPU) executing first instructions. Additionally or alternatively, the audio creation circuitry **120** of FIG. **2** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by (i) an Application Specific Integrated Circuit (ASIC) and/or (ii) a Field Programmable Gate Array (FPGA) structured and/or configured in response to execution of second instructions to perform operations corresponding to the first instructions. It should be understood that some or all of the circuitry of FIG. **2** may, thus, be instantiated at the same or different times. Some or all of the circuitry of FIG. **2** may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. **2** may be implemented by microprocessor circuitry executing instructions and/or FPGA circuitry performing operations to implement one or more virtual machines and/or containers. The example audio creation circuitry **120** of the illustrated example of FIG. **2** includes example source media accessor circuitry **205**, example media attribute detector circuitry **210**, example user input receiver circuitry **215**, example prompt generator circuitry **220**, example audio generator circuitry **225**, and example audio provider circuitry **230**.

[0033] The example source media accessor circuitry **205** of the illustrated example of FIG. **2** accesses a source media **110** for which audio tracks are to be generated. The example source media accessor circuitry **205** identifies a format of the source media **110** (e.g., .webm, .mpg, .mp2, .mpeg, .mpe, .mpv, .ogg, .mp4, .m4p, .m4v, .avi, .wmv, .mov, .qt, .flv, .swf, etc.). In some examples, the source media access circuitry **205** converts the source media **110** from a first format to a second format. In some examples, the source media **110** includes multiple files. In some examples, the source media access circuitry **110** combines multiple source media **110** files into a single file for processing. In some examples, the source media accessor circuitry **205** splits a source media **110** file into multiple files (e.g., a first file for audio and a second file for video). In some examples, the source media accessor circuitry **205** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0034] In some examples, the audio creation circuitry **120** includes means for accessing source media. For example, the means for accessing may be implemented by source media accessor circuitry **205**. In some examples, the source media accessor circuitry **205** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the source media accessor circuitry **205** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **702** of FIG. **7**. In some examples, the source media accessor circuitry **205** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the source media accessor circuitry **205** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the source media accessor circuitry **205** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0035] The example media attribute detector circuitry **210** of the illustrated example of FIG. **2** detects media attributes of the source media **110**. In some examples, the media attribute detector circuitry **210** executes a machine learning model to analyze the source media **110**. The example media attribute detector circuitry **210** identifies a variety of video characteristics (e.g., media attributes), including, but not limited to: whether, and where, the clip contains dialogue; color tone of the video; a type of activity being performed in the video; speed of action in the video; environment(s) detected in the video; person(s) detected in the video; number of people or activity detection; happiness detection; and type of video.

[0036] The example media attribute detector circuitry **210** performs voice isolation processing to identify segments of the source media **110** that contain voice. In some examples, the media attribute detector circuitry **210** identifies start times and end times for segments of the source media

110 containing voice. In some examples, the media attribute detector circuitry **210** generates a unit step function based on the identified segment(s). The unit step function is a function of time that has a value of zero for all values of time except during the time period corresponding to the source media **110** segments identified as containing voice, where the unit step function has a non-zero value (e.g., a value of 1).

[0037] In some examples, the media attribute detector circuitry **210** performs color analysis on the source media **110** to determine the color grading/warmth of the video. In those examples, the media attribute detector circuitry **210** uses the color tone of the source media **110** to derive a mood of the video. In some examples, the media attribute detector circuitry **210** generates graphs representing the color distribution of the source media **110**. These graphs can include, but are not limited to histogram graphs including for each of the red pixel values, green pixel values, and blue pixel values in the source media **110**, CIE Chromaticity graphs, waveform graphs, vector scope graphs, etc. The media attribute detector circuitry **210** analyzes the color grading of the source media **110** to determine whether the source media **110** has a bias towards colder (e.g., darker, moodier, scarier, etc.) tones or warmer (e.g., happy, fun, light, etc.) tones.

[0038] The example media attribute detector circuitry **210** uses facial recognition techniques and/or other object recognition techniques to determine the number of people that appear in a scene in the source media **110**. In some examples, the media attribute detector circuitry **210** identifies specific individuals that appear in the source media **110** using these techniques. In some examples, the media attribute detector circuitry **210** performs people detection using facial recognition and/or or other object recognition techniques. In some examples, the media attribute detection circuitry **210** detects environments in the source media **110**. Environment detection can include, for example, identifying the scene takes place by a waterfall or an ocean, which can be used to generate an audio track that is suitable for flowing water. In some examples, the media attribute detector circuitry **210** identifies the type of activity being performed. For example, the media attribute detector circuitry **210** can identify that a sport is being played and generates an audio track with a genre of music that matches that sport.

[0039] The example media attribute detector circuitry **210** generates values for music generation parameters based on the media attributes detected from the source media **110**. For example, the media attribute detector circuitry **210** may generate a value of 10 (on a scale of 1-10) for a happiness level parameter corresponding to a detected happiness level a source media **110** of a sports team winning a championship game. In some examples, media attribute parameters values are numbers within a defined range (e.g., a 0-10 scale, a 0-1 scale, etc.), numbers corresponding to an absolute value (e.g., the number of people in a video), and/or qualitative values (e.g., the type of activity being performed).

[0040] In some examples, the media attribute detection circuitry **210** identifies different scene segments in the source media **110** based on the detected media attributes. The media attribute detection circuitry **210** identifies changes in scenes by identifying changes in detected media attributes. For example, the media attribute detection circuitry **210** may use a detected abrupt change in the environment in the source media **110** to mark the end of a first scene segment and the beginning of a second scene segment. In some examples, the media attribute detector circuitry **210** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0041] In some examples, the audio creation circuitry **120** includes means for detecting media attributes. For example, the means for detecting may be implemented by media attribute detector circuitry **210**. In some examples, the media attribute detector circuitry **210** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the media attribute detector circuitry **210** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **704**, **706** of FIG. **7**. In some examples, the media attribute detector circuitry **210** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the media attribute detector circuitry **210** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the media attribute detector circuitry **210** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0042] The example user input receiver circuitry **215** of the illustrated example of FIG. **2** accesses user preferences for audio generation. User preferences include but are not limited to duration, background music priority, music preference, a preferred music generating machine learning model, weights and/or other instructions for use of the media attributes. In some examples, user preferences include a desired variable for the audio track to be generated (e.g., a specific genre of music to be used when a particular individual appears in the video). In some examples, the user input receiver circuitry **215** causes presentation of a user interface to allow a content creator **105** to input the user preferences.

[0043] In some examples, the user interface enables the content creator **105** to select user preferences from a pre-defined list (e.g., a list of music genres), adjust audio generation variables on a sliding scale (e.g., a desired bpm for the generated audio), and/or enter plain language instructions for audio generation. In some examples, the user input receiver circuitry **215** performs natural language comprehension to extract meaning from a plain language input from the content creator **105**. For example, the content creator **105** can input text stating a desired music genre be used for a specific person appearing in the source media **110** and the user input receiver circuitry **215** converts the input to variables to be used in generation of the audio track. In some examples, the user input receiver circuitry **215** enables a user to update user preferences for generation of an updated audio track based on the updated user preferences. In some examples, the user interface also enables the content creator **105** to set weighting values for corresponding attributes/preferences. For example, the user interface enables the user to place a high or low priority on the color tone of the video influencing the generation of the audio track. The user input receiver circuitry **215** accesses the weighting values set by the content creator **105** for use in audio generation. In some examples, the user input receiver circuitry **215** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0044] In some examples, the audio creation circuitry **120** includes means for accessing user preferences. For example, the means for accessing may be implemented by user input receiver circuitry **215**. In some examples, the user input receiver circuitry **215** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the user input receiver circuitry **215** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **708**, **710** of FIG. **7**. In some examples, the user input receiver circuitry **215** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the user input receiver circuitry **215** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the user input receiver circuitry **215** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0045] The example prompt generator circuitry **220** of the illustrated example of FIG. **2** generates prompts for a music generating machine learning model based on the video characteristics and the user preferences. The prompt generator circuitry **220** creates the prompt(s) based on a plurality of entries, some of the entries related to the user preferences and some of the entries related to the identified media attributes. In some examples, each entry includes an index to indicate whether the entry is related to a user preference or a media attribute, the variable the entry pertains to (e.g., a sub-index), and a value for the variable. In some examples, the prompt generator circuitry **220** implements a text management structure. The text management structure is a table including the plurality of entries. In some examples, prompt generator circuitry **220** executes a programming language to extract text from the entries to form the prompt of AI music generator. In some examples, the prompt generator circuitry **220** generates the prompt based on the specific music generating machine learning model to be used. For example, a first machine learning model may require a first input format/syntax and a second machine learning model may require a second input format/syntax. The prompt generator circuitry **220** generates the prompt to meet the format/syntax requirements of the machine learning model to be used. In some examples, the prompt generator circuitry **220** generates multiple prompts for a single source media **110**. In some examples, the prompt generator circuitry **220** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0046] In some examples, the audio creation circuitry **120** includes means for generating prompts. For example, the means for generating may be implemented by prompt generator circuitry **220**. In some examples, the prompt generator circuitry **220** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the prompt generator circuitry **220** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **712** of FIG. **7**. In some examples, the prompt generator circuitry **220** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the prompt generator circuitry **220** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the prompt generator circuitry **220** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0047] The example audio generator circuitry **225** of the illustrated example of FIG. **2** generates audio track(s) for the source media **110**. In some examples, the audio generator circuitry **225** executes a machine learning model to generate an audio track based on the prompt generated from the video characteristics and user preferences. In some examples, the audio generator circuitry **225** sends the prompt to an external music generating machine learning model. In some examples, the audio generator circuitry **225** executes a machine learning model selected by the user. In some examples, the audio generator circuitry **225** generates multiple audio tracks for the source media **110**. In some examples, the audio generator circuitry **225** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0048] In some examples, the audio creation circuitry **120** includes means for generating audio. For example, the means for generating may be implemented by audio generator circuitry **225.** In some examples, the audio generator circuitry **225** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the audio generator circuitry **225** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **714**, **716** of FIG. **7**. In some examples, the audio generator circuitry **225** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the audio generator circuitry **225** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the audio generator circuitry **225** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0049] The example audio provider circuitry **230** of the illustrated example of FIG. **2** provides the generated audio tracks to the content creator **105**. In some examples, the audio provider circuitry **230** provides the generated audio to a media creator application. In some examples, the audio provider circuitry **230** provides the audio track to the content creator **105** directly. In some examples, the audio provider circuitry **230** combines the generated audio track(s) with the source media **110** to generate a combined media and provides the combined media to the content creator **105**, directly or via the media creator application. In some examples, the audio provider circuitry **230** is instantiated by programmable circuitry executing source media accessor circuitry instructions and/or configured to perform operations such as those represented by the flowchart of FIG. **7**.

[0050] In some examples, the audio creation circuitry **120** includes means for providing audio tracks. For example, the means for providing may be implemented by audio provider circuitry **230**. In some examples, the audio provider circuitry **230** may be instantiated by programmable circuitry such as the example programmable circuitry **1112** of FIG. **11**. For instance, the audio provider circuitry **230** may be instantiated by the example microprocessor **1200** of FIG. **12** executing machine executable instructions such as those implemented by at least blocks **718** of FIG. **7**. In some examples, the audio provider circuitry **230** may be instantiated by hardware logic circuitry, which may be implemented by an ASIC, XPU, or the FPGA circuitry **1300** of FIG. **13** configured and/or structured to perform operations corresponding to the machine-readable instructions. Additionally or alternatively, the audio provider circuitry **230** may be instantiated by any other combination of hardware, software, and/or firmware. For example, the audio provider circuitry **230** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, an XPU, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) configured and/or structured to execute some or all of the machine-readable instructions and/or to perform some or all of the operations corresponding to the machine-readable instructions without executing software or firmware, but other structures are likewise appropriate.

[0051] While an example manner of implementing the audio creation circuitry **120** of FIG. **1** is illustrated in FIG. **2**, one or more of the elements, processes, and/or devices illustrated in FIG. **2** may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the example source media accessor circuitry **205**, the example media attribute detector circuitry **210**, the example user input receiver **215**, the example prompt generator circuitry **220**, the example audio generator circuitry **225**, the example audio provider circuitry **230**, and/or, more generally, the example audio creation circuitry **120** of FIG. **2**, may be implemented by hardware alone or by hardware in combination with software and/or firmware. Thus, for example, any of the example source media accessor circuitry **205**, the example media attribute detector circuitry **210**, the example user input receiver **215**, the example prompt generator circuitry **220**, the example audio generator circuitry **225**, the example audio provider circuitry **230**, and/or, more generally, the example audio creation circuitry **120**, could be implemented by programmable circuitry in combination with machine-readable instructions (e.g., firmware or software), processor circuitry, analog circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), ASIC(s), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as FPGAs. Further still, the example audio creation circuitry **120** of FIG. **2** may include one or more elements, processes, and/or devices in addition to, or instead of, those illustrated in FIG. **2**, and/or may include more than one of any or all of the illustrated elements, processes and devices.

[0052] A flowchart representative of example machine-readable instructions, which may be executed by programmable circuitry to implement and/or instantiate the audio creation circuitry **120** of FIG. **2** and/or representative of example operations which may be performed by programmable circuitry to implement and/or instantiate the audio creation circuitry **120** of FIG. **2**, is shown in FIG. **7**. The machine-readable instructions may be one or more executable programs or portion(s) of one or more executable programs for execution by programmable circuitry such as the programmable circuitry **1112** shown in the example processor platform **1100** discussed below in connection with FIG. **11** and/or may be one or more function(s) or portion(s) of functions to be performed by the example programmable circuitry (e.g., an FPGA) discussed below in connection with FIGS. **12** and/or **13**. In some examples, the machine-readable instructions cause an operation, a task, etc., to be carried out and/or performed in an automated manner in the real world. As used herein, “automated” means without human involvement.

[0053] The program may be embodied in instructions (e.g., software and/or firmware) stored on one or more non-transitory computer-readable and/or machine-readable storage medium such as cache memory, a magnetic-storage device or disk (e.g., a floppy disk, a Hard Disk Drive (HDD), etc.), an optical-storage device or disk (e.g., a Blu-ray disk, a Compact Disk (CD), a Digital Versatile Disk (DVD), etc.), a Redundant Array of Independent Disks (RAID), a register, ROM, a solid-state drive (SSD), SSD memory, non-volatile memory (e.g., electrically erasable programmable read-only memory (EEPROM), flash memory, etc.), volatile memory (e.g., Random Access Memory (RAM) of any type, etc.), and/or any other storage device or storage disk. The instructions of the non-transitory computer-readable and/or machine-readable medium may program and/or be executed by programmable circuitry located in one or more hardware devices, but the entire program and/or parts thereof could alternatively be executed and/or instantiated by one or more hardware devices other than the programmable circuitry and/or embodied in dedicated hardware. The machine-readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a human and/or machine user) or an intermediate client hardware device gateway (e.g., a radio access network (RAN)) that may facilitate communication between a server and an endpoint client hardware device. Similarly, the non-transitory computer-readable storage medium may include one or more mediums. Further, although the example program is described with reference to the flowchart illustrated in FIG. **7**, many other methods of implementing the example audio creation circuitry **120** may alternatively be used. For example, the order of execution of the blocks of the flowchart may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks of the flow chart may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The programmable circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core CPU), a multi-core processor (e.g., a multi-core CPU, an XPU, etc.)). For example, the programmable circuitry may be a CPU and/or an FPGA located in the same package (e.g., the same integrated circuit (IC) package or in two or more separate housings), one or more processors in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, etc., and/or any combination(s) thereof.

[0054] The machine-readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine-readable instructions as described herein may be stored as data (e.g., computer-readable data, machine-readable data, one or more bits (e.g., one or more computer-readable bits, one or more machine-readable bits, etc.), a bitstream (e.g., a computer-readable bitstream, a machine-readable bitstream, etc.), etc.) or a data structure (e.g., as portion(s) of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine-readable instructions may be fragmented and stored on one or more storage devices, disks and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine-readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine-readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of computer-executable and/or machine executable instructions that implement one or more functions and/or operations that may together form a program such as that described herein.

[0055] In another example, the machine-readable instructions may be stored in a state in which they may be read by programmable circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine-readable instructions on a particular computing device or other device. In another example, the machine-readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine-readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine-readable, computer-readable and/or machine-readable media, as used herein, may include instructions and/or program(s) regardless of the particular format or state of the machine-readable instructions and/or program(s).

[0056] The machine-readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine-readable instructions may be represented using any of the following languages: C, C++, Java, C #, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

[0057] As mentioned above, the example operations of FIG. **7** may be implemented using executable instructions (e.g., computer-readable and/or machine-readable instructions) stored on one or more non-transitory computer-readable and/or machine-readable media. As used herein, the terms non-transitory computer-readable medium, non-transitory computer-readable storage medium, non-transitory machine-readable medium, and/or non-transitory machine-readable storage medium are expressly defined to include any type of computer-readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media. Examples of such non-transitory computer-readable medium, non-transitory computer-readable storage medium, non-transitory machine-readable medium, and/or non-transitory machine-readable storage medium include optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory (ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms “non-transitory computer-readable storage device” and “non-transitory machine-readable storage device” are defined to include any physical (mechanical, magnetic and/or electrical) hardware to retain information for a time period, but to exclude propagating signals and to exclude transmission media. Examples of non-transitory computer-readable storage devices and/or non-transitory machine-readable storage devices include random access memory of any type, read only memory of any type, solid state memory, flash memory, optical discs, magnetic disks, disk drives, and/or redundant array of independent disks (RAID) systems. As used herein, the term “device” refers to physical structure such as mechanical and/or electrical equipment, hardware, and/or circuitry that may or may not be configured by computer-readable instructions, machine-readable instructions, etc., and/or manufactured to execute computer-readable instructions, machine-readable instructions, etc.

[0058] FIG. **3** is a waveform graph **300** illustrating example voice audio isolation processing performed by the audio creation circuitry **120** of FIG. **1**. The waveform graph **300** includes a time axis **302** and an amplitude axis **304**. The waveform graph **300** includes a first audio track **310**, a second audio track **320**, and a unit step function **330**. The first audio track **310** is a source audio track captured from the source media **110**. In some examples, the audio creation circuitry **120** performs voice isolation processing to isolate a portion of the audio track that contains voice. The second audio track **320** represents the first audio track **310** with portions not containing voice removed. In some examples, the audio creation circuitry **120** identifies a start time **322** and an end time **324** for a segment **326** of the second audio track **320** containing voice. In some examples, the audio creation circuitry **120** identifies a plurality of segments **326** for the second audio track containing voice and identifies a plurality of corresponding start times **322** and end times **324**.

[0059] In some examples, the audio creation circuitry **120** generates a unit step function **330** based on the identified segment(s) **326**. The unit step function **330** is a function of time that has a value of zero for all values of time except between start time(s) **322** and corresponding end time(s) **324**, where the unit step function **330** has a non-zero value (e.g., a value of 1). In some examples, the unit step function **330** may be represented by a data structure that encodes start times **322** and stop times **324**. In some examples, the audio creation circuitry **120** generates audio based on the unit step function **330**. For example, the audio creation circuitry **120** reduces the dynamic range of generated audio (e.g., does not include drum beats or guitar riffs) for values of time where the unit step function has a non-zero value.

[0060] FIG. **4** is an example cold-lighting video **400** analyzed by the audio creation circuitry **120** of FIG. **1**. In the illustrated example of FIG. **4**, the audio creation circuitry **120** generates a histogram graph **410** including a red color histogram **420**, a green color histogram **430**, and a blue color histogram **440**. The x axis of the histograms **420**, **430**, and **440** represents the pixel intensity and the y axis represents the number of pixels. Each pixel in the cold-lighting video **400** can be represented by a brightness value for each of the colors red, green, and blue. The audio creation circuitry **120** generates the histograms **420**, **430**, **440** by determining the red, green, and blue pixel intensities for each pixel in the cold-lighting video **400** and using that plotting the number of pixels corresponding to each pixel intensity. In the illustrated example of FIG. **4**, the blue histogram **440** has a spike **442**, at a pixel intensity of about 1000, indicating there is a large amount of blue in the cold-lighting video **400**. In some examples, the audio creation circuitry **120** generates other graphs representing the color distribution of the cold-lighting video **400**. For example, the audio creation circuitry **120** generates CIE Chromaticity graphs, waveform graphs, vector scope graphs, etc. The audio creation circuitry **120** analyzes the color grading of the source media **110** to determine whether the source media **110** has a bias towards colder (e.g., darker, moodier, scarier, etc.) tones or warmer (e.g., happy, fun, light, etc.) tones. In the illustrated example of FIG. **4**, the audio creation circuitry **120** identifies the cold-lighting video **400** as having a colder mood. The audio creation circuitry **120** generates colder audio (e.g., an instrumental, music with melancholy lyrics, etc.) based on the identified colder mood.

[0061] FIG. **5** is an example warm-lighting video **500** analyzed by the audio creation circuitry **120** of FIG. **1**. In the illustrated example of FIG. **5**, the audio creation circuitry **120** generates a histogram graph **510** including a red color histogram **520**, a green color histogram **530**, and a blue color histogram **540**. The x axis of the histograms **520**, **530**, and **540** represents the pixel intensity and the y axis represents the number of pixels. The audio creation circuitry **120** generates the histograms **520**, **530**, **540** by determining the red, green, and blue pixel intensities for each pixel in the warm-lighting video **500** and using that plotting the number of pixels corresponding to each pixel intensity. In the illustrated example of FIG. **5**, the red histogram **520** has a spike **522**, at a pixel intensity of about **1000**, indicating there is a large amount of red in the warm-lighting video **500**. In some examples, the audio creation circuitry **120** generates other graphs representing the color distribution of the warm-lighting video **500**. For example, the audio creation circuitry **120** generates CIE Chromaticity graphs, waveform graphs, vector scope graphs, etc. In the illustrated example of FIG. **5**, the audio creation circuitry **120** identifies the warm-lighting video **500** as having a warmer mood. The audio creation circuitry **120** generates happier audio (e.g., music positive lyrics, pop genre, etc.) based on the identified warmer mood.

[0062] FIG. **6** illustrates person and/or scene detection as performed by the audio creation circuitry **120** of FIG. **1**. FIG. **6** includes a plurality of frames **600** from an example source media **110**. A first portion of the plurality of frames occur in a first environment **604** and a second environment **606**. The first environment **604** is a walking trail in a nature reserve and the second environment **606** is a miniature golf course. The audio creation circuitry **120** identifies the environments **604**, **606** and generates audio tracks appropriate for the detected environments **604**, **606**. For example, the audio creation circuitry **120** may generate an audio track that evokes a feeling of awe or wonder for the walking trail environment. In some examples, the audio creation circuitry **120** performs person detection to determine a number of people in a scene. In some examples, the audio creation circuitry **120** generates audio tracks based on the determined number of people in a scene. For example, detecting only one person in a scene may lead the audio creation circuitry **120** to generate an audio track that evokes a feeling of isolation, depending on the other input variables from the user preferences and the video analysis. In some examples, the audio creation circuitry **120** identifies specific persons across multiple scenes. In the illustrated example of FIG. **6**, the audio creation circuitry **120** identified a person **602** that appears throughout the plurality of frames **600**. In some examples, the audio creation circuitry **120** generates audio with an emphasis on audio variables for identified persons. For example, the audio creation circuitry **120** may generate an audio track that uses or emphasizes a guitar track every time person **602** appears in the video. In some examples, the content creator **105** can set preferences for music generation variables for specific individuals.

[0063] FIG. **7** is a flowchart representative of example machine-

[0064] readable instructions and/or example operations **700** that may be executed, instantiated, and/or performed by programmable circuitry to generate audio tracks for source media. The example machine-readable instructions and/or the example operations **700** of FIG. **7** begin at block **702**, at which the source media accessor circuitry **205** accesses source media **110**. The source media **110** may be any one of a variety of formats (e.g., .webm, .mpg, .mp2, .mpeg, .mpe, .mpv, .ogg, .mp4, .m4p, .m4v, .avi, .wmv, .mov, .qt, .flv, .swf, etc.). In some examples, the source media access circuitry **205** converts the source media **110** from a first format to a second format. In some examples, the source media **110** includes multiple files. In some examples, the source media access circuitry **110** combines source media **110** files into a single file for processing. In other examples, multiple source media **110** files are processed separately.

[0065] At block **704**, the media attribute detector circuitry **210** analyzes the source media **110** to detect a media attribute of the source media **110**. The media attribute detector circuitry **210** can detect a variety of media attributes of the source media **110**, such as whether, and where, the clip contains dialogue; color tone of the video; a type of activity being performed in the video; speed of action in the video; environment(s) detected in the video; person(s) detected in the video; number of people or activity detection; happiness detection; type of video, etc. In some examples, the media attribute detector circuitry **210** executes a machine learning model to analyze the source media **110**. In some examples, the media attribute detector circuitry **210** performs voice isolation processing of the source media **110** to identify dialogue. In some examples, to determine the color grading/warmth of the source media **110**, the media attribute detector circuitry **210** performs color analysis. The media attribute detector circuitry **210** may use facial recognition techniques and/or other object recognition techniques to identify specific individuals that appear in the source media **110**. However, any other attributes detection technique(s) may additionally or alternatively be used.

[0066] In some examples, the media attribute detector circuitry **210** generates values for media attribute parameters corresponding to media attributes detected from the source media **110**. For example, the media attribute detector circuitry **210** may generate a value of ten (on a scale of zero to ten) for a happiness level parameter corresponding to a detected happiness level a source media **110** of a sports team winning a championship game. In some examples, media attribute parameters values are numbers within a defined range, numbers corresponding to an absolute value, and/or qualitative values.

[0067] In some examples, the media attribute detection circuitry **210** identifies different scene segments in the source media **110**. The media attribute detection circuitry **210** identifies changes in scenes by identifying changes in detected media attributes. For example, the media attribute detection circuitry **210** analyzes the source media **110** to identify portions of the source media **110** with different happiness levels, environments, and amount of dialogue and segments the source media **110** into different scenes (e.g., a first scene segment depicting a first scene such as a child's birthday party and a second scene segment depicting a second scene such as a serious conversation between the child's parents) to generate an audio track(s) that are appropriate for each scene segment. In some examples, the media attribute circuitry **210** determines whether the difference in media attributes between two scenes is significant enough to warrant switching to a new style of music and/or distinct audio track or if the requirements of both scenes can be met by adjusting one or more variables of the audio track (e.g., lowering the volume of the audio track in the second scene segment).

[0068] At decision block **706**, the media attribute detector circuitry **210** determines whether there are additional media attributes to detect. If there are more attributes to detect (e.g., block **706** returns a result of “YES”), the example method **700** returns to block **704** and the media attribute detector circuitry **210** detects another media attribute. If there are no more media attributes to detect (e.g., block **706** returns a result of “NO”), the method proceeds to block **708**, at which the user input receiver circuitry **215** accesses user preference selections. User preferences include but are not limited to: duration; background music priority; music preference; a preferred music generating machine learning model; weights, values, and/or other instructions for use of the media attributes.

[0069] In some examples, user preferences include a desired variable for the audio track to be generated (e.g., a specific genre of music to be used when a particular individual appears in the video). In some examples, the user input receiver circuitry **215** enables the user to select user preferences from a pre-defined list (e.g., a list of music genres). In some examples, the user input receiver circuitry **215** enables the user to communicate user preferences in plain language user input and processes the user input using natural language comprehension to identify user preference parameters. At block **710**, the user input receiver circuitry **215** accesses weighting values for corresponding attributes/preferences. For example, the user input receiver circuitry **215** enables the user to place a high or low priority on the color tone of the video influencing the generation of the audio track. As another example, the user input receiver circuitry **215** enables the user to place a weight on the importance of a user preference corresponding to a minimum length of time for consistent music to be played.

[0070] At block **712**, the prompt generator circuitry **220** constructs a prompt for a machine learning model based at least in part on detected media attributes, user preferences, and weighting values. The prompt generator circuitry **220** creates the prompt(s) based on a plurality of entries, some of the entries related to the user preferences and some of the entries related to the identified media attributes. In some examples, each entry includes an index to indicate whether the entry is related to a user preference or a media attribute, the variable the entry pertains to, and a value for the variable. In some examples, the prompt generator circuitry **220** executes a programming language to extract text from the plurality of entries to form one or two sentences as the prompt of an AI music generator.

[0071] At block **714**, the audio generator circuitry **225** generates audio track(s) based on the prompt. In some examples, the audio generator circuitry **225** executes a machine learning model to generate the audio. In some examples, the audio generator circuitry **225** sends the prompt to an external music generating machine learning model. In some examples, the audio generator circuitry **225** executes a machine learning model selected by the user. At block **716**, the audio generator circuitry **225** determines whether there are additional audio(s) to create. If there are (e.g., block **706** returns a result of “YES”), the method returns to block **714** and the audio generator circuitry **225** generates more audio(s). If there are not (e.g., block **706** returns a result of “NO”), the method proceeds to block **718**, at which the audio provider circuitry **230** provides the generated audio to a media creator application. In some examples, the audio provider circuitry **230** provides the audio track to the content creator **105** directly. In some examples, the audio provider circuitry **230** combines the generated audio track(s) with the source media **110** to generate a combined media and provides the combined media to the content creator **105**, directly or via the media creator application. Then, the example method **700** terminates.

[0072] FIG. **8** is a table illustrating an example text management data structure **800** generated by the audio creation circuitry **120** of FIG. **1**. The example text management data structure **800** is a table including example user preference entries **810** and example video analysis result entries **812**. In some examples, In the illustrated example of FIG. **8**, each entry **810**, **812** in the example text management data structure **800** includes a text management structure index **802** indicating whether the entry is derived from user preference input variables or a video analysis result, a text management structure sub-index **804** (e.g., the specific variable the entry is directed to), and a text management structure value **806** indicating the value for the variable. In some examples, the text management structure **800** includes additional or alternative information in each entry **810**, **812** (e.g., a weight assigned to the entry **810**, **812**). In some examples, the text management structure **800** includes information and/or is formatted based on the music-generating machine learning model used by the audio creation circuitry **120**. The example text management structure **800** is in an example format, however, other text management structure **800** formats may additionally or alternatively be used. The example user preference entries **810** correspond to a user input of 10 s duration and pop music. The example video analysis result entries **812** correspond to a happy scene. In some examples, audio creation circuitry **120** executes a programming language to extract the text to form the prompt for the music-generating machine learning model. The programming language, sequentially or in parallel, extracts the data text from the text management data structure **800** to generate the prompt.

[0073] FIG. **9** is a block diagram illustrating an example workload schedule **900** implemented by the audio creation circuitry **120** on an example system **910**. The example system **910** includes an example neural processing unit (NPU) **920**, an example central processing unit (CPU) **930**, and an example graphics processing unit (GPU) **940**. In some examples, the audio creation circuitry **120** detects the configuration of a system and submits the workload(s) to the appropriate hardware engine to maximize efficiency. In the illustrated example of FIG. **9**, the audio creation circuitry **120** submits video analysis to be performed by the NPU **920**, prompt generation (e.g., text management) to be performed by the CPU **930**, and music generation to be performed by the GPU **940**.

[0074] In some examples, the audio creation circuitry **120** causes parallel processing of the workloads. In the illustrated example of FIG. **9**, the NPU **920** performs a first video analysis workload **922** for a first scene, a second video analysis workload **924** for a second scene, and a third video analysis workload **926** for a third scene. The example NPU **920** performs the workloads **922**, **924**, **926** at least partially in parallel. As used herein, performance of workloads "at least partially in parallel" means that performance of a second workload may begin before a first workload is completed, but after the first workload was started. In some examples, the performance of the workloads may be performed in parallel, with multiple workloads being started at substantially the same time. The example CPU **930** performs a first prompt generation workload **932**, a second prompt generation workload **934**, and a third prompt generation workload **936**. The example CPU **930** performs the workloads **932**, **934**, **936** at least partially in parallel. The example GPU **940** performs a first music generation workload **942**, a second music generation workload **944**, and a third music generation workload **946**. The example GPU **940** performs the workloads **942**, **944**, **946** at least partially in parallel. In some examples, the distribution of workload between the NPU **920**, the CPU **930**, and the GPU **940** is different than as shown in FIG. **9**. For example, the NPU **920** and the GPU **940** may share responsibility to complete the analysis workloads **922**, **924**, **926**.

[0075] In some examples, the system **910** includes a subset of the NPU **920**, the CPU **930**, and the GPU **940**. For example, the system **910** may only include the CPU **930**. In that example, the workloads associated with the NPU **920**, and the GPU **940** of the example system **910** would be completed by the CPU **930**. In other examples, the system **910** includes additional XPUs (e.g., a DPU).

[0076] FIG. **10** is an example user interface **1000** implemented by the audio creation circuitry **120** of FIG. **1**. In some examples, the example user interface **1000** may be referred to as a generative music prioritization console. The example user interface **1000** enables the content creator **105** to input user preferences for use in audio generation. The example user interface includes an example scene selection panel **1002**, and example track duration panel **1004**, an example genre selection panel **1006**, an example input variable prioritization panel **1008**, an example input control panel **1010**, an example output effect panel **1012**, and an example create button **1014**. The scene identifier **1002** displays information identifying the scene the inputs of the content creator **105** will affect music generation for and enables the content creator **105** to select which scene to set preferences for. The user interface **1000** enables the content creator **105** to edit several recordings (from various events) together with minimal effort without having to edit one at a time manually. The scene selection panel **1002** enables the content creator **105** to enter user preferences for the entire video sequence, user preferences for each scene, or even clip level user preferences to direct the audio generation. In some examples, the user interface **1000** also displays an image from the corresponding scene to the content creator **105**. The track duration panel **1004** enables the content creator **105** to set a desired track duration for audio generation. In the illustrated example of FIG. **10**, the content creator **105** set a minimum track duration limit of thirty seconds and a maximum track duration limit of three minutes. The genre selection panel **1006** enables the content creator **105** to select a desired genre of music for the selected scene. The example genre selection panel **1006** of FIG. **10** indicates the content creator **105** has selected the genre "pop." The input variable prioritization panel **1008** enables the content creator **105** to apply a weight to the input variables for music generation. In the illustrated example of FIG. **10**, the environment awareness variable is set to the highest weight of the input variables, while the dialogue priority is set to the lowest weight.

[0077] Each of the video analysis variables has its own unique set of control variables. The input control panel **1010** enables the content creator **105** to select a variable and the output effect panel **1012** enables the content creator **105** to select a control for the selected variable. In the illustrated example of FIG. **10**, the dialogue priority variable has two options: "Dynamic Range Cut" or "Amplitude Cut." The output control panel of FIG. **10** indicates that the content creator has selected the "Dynamic Range Cut" option. Thus, the dynamic range of the music would be reduced when voice is detected in the source media **110**. In some examples, the user interface **1000** enables the content creator **105** to adjust user preferences for iterative adjustments to the generated music. While the video analysis and user inputs would create an initial set of recommendations and music, the content creator **105** can use the selection tools to adjust the user preferences for music generation by adjusting the user preferences accordingly to increase or decrease a particular variable's effect on the background music generative process.
[0078] FIG. **11** is a block diagram of an example programmable circuitry platform **1100** structured to execute and/or instantiate the example machine-readable instructions and/or the example operations of FIG. **7** to implement the audio creation circuitry **120** of FIG. **2**. The programmable circuitry platform **1100** can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), a mobile device (e.g., a cell phone, a smart phone, a tablet such as an iPad™), an Internet appliance, a gaming console, a headset (e.g., an augmented reality (AR) headset, a virtual reality (VR) headset, etc.) or other wearable device, or any other type of computing and/or electronic device.
[0079] The programmable circuitry platform **1100** of the illustrated example includes programmable circuitry **1112**. The programmable circuitry **1112** of the illustrated example is hardware. For example, the programmable circuitry **1112** can be implemented by one or more integrated circuits, logic circuits, FPGAs, microprocessors, CPUs, GPUs, DSPs, and/or microcontrollers from any desired family or manufacturer. The programmable circuitry **1112** may be implemented by one or more semiconductor based (e.g., silicon based) devices. In this example, the programmable circuitry **1112** implements the example source media accessor circuitry **205**, the example media attribute detector circuitry **210**, the example input receiver circuitry **215**, the example prompt generator circuitry **220**, the example audio generator circuitry **225**, and the example audio provider circuitry **230**.
[0080] The programmable circuitry **1112** of the illustrated example includes a local memory **1113** (e.g., a cache, registers, etc.). The programmable circuitry **1112** of the illustrated example is in communication with main memory **1114**, **1116**, which includes a volatile memory **1114** and a non-volatile memory **1116**, by a bus **1118**. The volatile memory **1114** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory **1116** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1114**, **1116** of the illustrated example is controlled by a memory controller **1117**. In some examples, the memory controller **1117** may be implemented by one or more integrated circuits, logic circuits, microcontrollers from any desired family or manufacturer, or any other type of circuitry to manage the flow of data going to and from the main memory **1114**, **1116**.
[0081] The programmable circuitry platform **1100** of the illustrated example also includes interface circuitry **1120**. The interface circuitry **1120** may be implemented by hardware in accordance with any type of interface standard, such as an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.
[0082] In the illustrated example, one or more input devices **1122** are connected to the interface circuitry **1120**. The input device(s) **1122** permit(s) a user (e.g., a human user, a machine user, etc.) to enter data and/or commands into the programmable circuitry **1112**. The input device(s) **1122** can

be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a trackpad, a trackball, an isopoint device, and/or a voice recognition system.

[0083] One or more output devices **1124** are also connected to the interface circuitry **1120** of the illustrated example. The output device(s) **1124** can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube (CRT) display, an in-place switching (IPS) display, a touchscreen, etc.), a tactile output device, a printer, and/or speaker. The interface circuitry **1120** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip, and/or graphics processor circuitry such as a GPU.

[0084] The interface circuitry **1120** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) by a network **1126**. The communication can be by, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a beyond-line-of-sight wireless system, a line-of-sight wireless system, a cellular telephone system, an optical connection, etc.

[0085] The programmable circuitry platform **1100** of the illustrated example also includes one or more mass storage discs or devices **1128** to store firmware, software, and/or data. Examples of such mass storage discs or devices **1128** include magnetic storage devices (e.g., floppy disk, drives, HDDs, etc.), optical storage devices (e.g., Blu-ray disks, CDs, DVDs, etc.), RAID systems, and/or solid-state storage discs or devices such as flash memory devices and/or SSDs.

[0086] The machine-readable instructions **1132**, which may be implemented by the machine-readable instructions of FIG. **7**, may be stored in the mass storage device **1128**, in the volatile memory **1114**, in the non-volatile memory **1116**, and/or on at least one non-transitory computer-readable storage medium such as a CD or DVD which may be removable.

[0087] FIG. **12** is a block diagram of an example implementation of the programmable circuitry **1112** of FIG. **11**. In this example, the programmable circuitry **1112** of FIG. **11** is implemented by a microprocessor **1200**. For example, the microprocessor **1200** may be a general-purpose microprocessor (e.g., general-purpose microprocessor circuitry). The microprocessor **1200** executes some or all of the machine-readable instructions of the flowcharts of FIG. **7** to effectively instantiate the circuitry of FIG. **2** as logic circuits to perform operations corresponding to those machine-readable instructions. In some such examples, the circuitry of FIG. **2** is instantiated by the hardware circuits of the microprocessor **1200** in combination with the machine-readable instructions. For example, the microprocessor **1200** may be implemented by multi-core hardware circuitry such as a CPU, a DSP, a GPU, an XPU, etc. Although it may include any number of example cores **1202** (e.g., 1 core), the microprocessor **1200** of this example is a multi-core semiconductor device including N cores. The cores **1202** of the microprocessor **1200** may operate independently or may cooperate to execute machine-readable instructions. For example, machine code corresponding to a firmware program, an embedded software program, or a software program may be executed by one of the cores **1202** or may be executed by multiple ones of the cores **1202** at the same or different times. In some examples, the machine code corresponding to the firmware program, the embedded software program, or the software program is split into threads and executed in parallel by two or more of the cores **1202**. The software program may correspond to a portion or all of the machine-readable instructions and/or operations represented by the flowcharts of FIG. **7**.

[0088] The cores **1202** may communicate by a first example bus **1204**. In some examples, the first bus **1204** may be implemented by a communication bus to effectuate communication associated with one(s) of the cores **1202**. For example, the first bus **1204** may be implemented by at least one of an Inter-Integrated Circuit (I2C) bus, a Serial Peripheral Interface (SPI) bus, a PCI bus, or a

PCIe bus. Additionally or alternatively, the first bus **1204** may be implemented by any other type of computing or electrical bus. The cores **1202** may obtain data, instructions, and/or signals from one or more external devices by example interface circuitry **1206**. The cores **1202** may output data, instructions, and/or signals to the one or more external devices by the interface circuitry **1206**. Although the cores **1202** of this example include example local memory **1220** (e.g., Level 1 (L1) cache that may be split into an L1 data cache and an L1 instruction cache), the microprocessor **1200** also includes example shared memory **1210** that may be shared by the cores (e.g., Level 2 (L2 cache)) for high-speed access to data and/or instructions. Data and/or instructions may be transferred (e.g., shared) by writing to and/or reading from the shared memory **1210**. The local memory **1220** of each of the cores **1202** and the shared memory **1210** may be part of a hierarchy of storage devices including multiple levels of cache memory and the main memory (e.g., the main memory **1114**, **1116** of FIG. **11**). Typically, higher levels of memory in the hierarchy exhibit lower access time and have smaller storage capacity than lower levels of memory. Changes in the various levels of the cache hierarchy are managed (e.g., coordinated) by a cache coherency policy.

[0089] Each core **1202** may be referred to as a CPU, DSP, GPU, etc., or any other type of hardware circuitry. Each core **1202** includes control unit circuitry **1214**, arithmetic and logic (AL) circuitry (sometimes referred to as an ALU) **1216**, a plurality of registers **1218**, the local memory **1220**, and a second example bus **1222**. Other structures may be present. For example, each core **1202** may include vector unit circuitry, single instruction multiple data (SIMD) unit circuitry, load/store unit (LSU) circuitry, branch/jump unit circuitry, floating-point unit (FPU) circuitry, etc. The control unit circuitry **1214** includes semiconductor-based circuits structured to control (e.g., coordinate) data movement within the corresponding core **1202**. The AL circuitry **1216** includes semiconductor-based circuits structured to perform one or more mathematic and/or logic operations on the data within the corresponding core **1202**. The AL circuitry **1216** of some examples performs integer based operations. In other examples, the AL circuitry **1216** also performs floating-point operations. In yet other examples, the AL circuitry **1216** may include first AL circuitry that performs integer-based operations and second AL circuitry that performs floating-point operations. In some examples, the AL circuitry **1216** may be referred to as an Arithmetic Logic Unit (ALU).

[0090] The registers **1218** are semiconductor-based structures to store data and/or instructions such as results of one or more of the operations performed by the AL circuitry **1216** of the corresponding core **1202**. For example, the registers **1218** may include vector register(s), SIMD register(s), general-purpose register(s), flag register(s), segment register(s), machine-specific register(s), instruction pointer register(s), control register(s), debug register(s), memory management register(s), machine check register(s), etc. The registers **1218** may be arranged in a bank as shown in FIG. **12**. Alternatively, the registers **1218** may be organized in any other arrangement, format, or structure, such as by being distributed throughout the core **1202** to shorten access time. The second bus **1222** may be implemented by at least one of an I2C bus, a SPI bus, a PCI bus, or a PCIe bus.

[0091] Each core **1202** and/or, more generally, the microprocessor **1200** may include additional and/or alternate structures to those shown and described above. For example, one or more clock circuits, one or more power supplies, one or more power gates, one or more cache home agents (CHAs), one or more converged/common mesh stops (CMSs), one or more shifters (e.g., barrel shifter(s)) and/or other circuitry may be present. The microprocessor **1200** is a semiconductor device fabricated to include many transistors interconnected to implement the structures described above in one or more integrated circuits (ICs) contained in one or more packages.

[0092] The microprocessor **1200** may include and/or cooperate with one or more accelerators (e.g., acceleration circuitry, hardware accelerators, etc.). In some examples, accelerators are implemented by logic circuitry to perform certain tasks more quickly and/or efficiently than can be done by a general-purpose processor. Examples of accelerators include ASICs and FPGAs such as those discussed herein. A GPU, DSP and/or other programmable device can also be an accelerator. Accelerators may be on-board the microprocessor **1200**, in the same chip package as the

microprocessor **1200** and/or in one or more separate packages from the microprocessor **1200**.
[0093] FIG. **13** is a block diagram of another example implementation of the programmable circuitry **1112** of FIG. **11**. In this example, the programmable circuitry **1112** is implemented by FPGA circuitry **1300**. For example, the FPGA circuitry **1300** may be implemented by an FPGA. The FPGA circuitry **1300** can be used, for example, to perform operations that could otherwise be performed by the example microprocessor **1200** of FIG. **12** executing corresponding machine-readable instructions. However, once configured, the FPGA circuitry **1300** instantiates the operations and/or functions corresponding to the machine-readable instructions in hardware and, thus, can often execute the operations/functions faster than they could be performed by a general-purpose microprocessor executing the corresponding software.
[0094] More specifically, in contrast to the microprocessor **1200** of FIG. **12** described above (which is a general purpose device that may be programmed to execute some or all of the machine-readable instructions represented by the flowchart of FIG. **7** but whose interconnections and logic circuitry are fixed once fabricated), the FPGA circuitry **1300** of the example of FIG. **13** includes interconnections and logic circuitry that may be configured, structured, programmed, and/or interconnected in different ways after fabrication to instantiate, for example, some or all of the operations/functions corresponding to the machine-readable instructions represented by the flowchart of FIG. **7**. In particular, the FPGA circuitry **1300** may be thought of as an array of logic gates, interconnections, and switches. The switches can be programmed to change how the logic gates are interconnected by the interconnections, effectively forming one or more dedicated logic circuits (unless and until the FPGA circuitry **1300** is reprogrammed). The configured logic circuits enable the logic gates to cooperate in different ways to perform different operations on data received by input circuitry. Those operations may correspond to some or all of the instructions (e.g., the software and/or firmware) represented by the flowchart of FIG. **7**. As such, the FPGA circuitry **1300** may be configured and/or structured to effectively instantiate some or all of the operations/functions corresponding to the machine-readable instructions of the flowchart of FIG. **7** as dedicated logic circuits to perform the operations/functions corresponding to those software instructions in a dedicated manner analogous to an ASIC. Therefore, the FPGA circuitry **1300** may perform the operations/functions corresponding to the some or all of the machine-readable instructions of FIG. **7** faster than the general-purpose microprocessor can execute the same.
[0095] In the example of FIG. **13**, the FPGA circuitry **1300** is configured and/or structured in response to being programmed (and/or reprogrammed one or more times) based on a binary file. In some examples, the binary file may be compiled and/or generated based on instructions in a hardware description language (HDL) such as Lucid, Very High Speed Integrated Circuits (VHSIC) Hardware Description Language (VHDL), or Verilog. For example, a user (e.g., a human user, a machine user, etc.) may write code or a program corresponding to one or more operations/functions in an HDL; the code/program may be translated into a low-level language as needed; and the code/program (e.g., the code/program in the low-level language) may be converted (e.g., by a compiler, a software application, etc.) into the binary file. In some examples, the FPGA circuitry **1300** of FIG. **13** may access and/or load the binary file to cause the FPGA circuitry **1300** of FIG. **13** to be configured and/or structured to perform the one or more operations/functions. For example, the binary file may be implemented by a bit stream (e.g., one or more computer-readable bits, one or more machine-readable bits, etc.), data (e.g., computer-readable data, machine-readable data, etc.), and/or machine-readable instructions accessible to the FPGA circuitry **1300** of FIG. **13** to cause configuration and/or structuring of the FPGA circuitry **1300** of FIG. **13**, or portion(s) thereof.
[0096] In some examples, the binary file is compiled, generated, transformed, and/or otherwise output from a uniform software platform utilized to program FPGAs. For example, the uniform software platform may translate first instructions (e.g., code or a program) that correspond to one or more operations/functions in a high-level language (e.g., C, C++, Python, etc.) into second

instructions that correspond to the one or more operations/functions in an HDL. In some such examples, the binary file is compiled, generated, and/or otherwise output from the uniform software platform based on the second instructions. In some examples, the FPGA circuitry **1300** of FIG. **13** may access and/or load the binary file to cause the FPGA circuitry **1300** of FIG. **13** to be configured and/or structured to perform the one or more operations/functions. For example, the binary file may be implemented by a bit stream (e.g., one or more computer-readable bits, one or more machine-readable bits, etc.), data (e.g., computer-readable data, machine-readable data, etc.), and/or machine-readable instructions accessible to the FPGA circuitry **1300** of FIG. **13** to cause configuration and/or structuring of the FPGA circuitry **1300** of FIG. **13**, or portion(s) thereof.

[0097] The FPGA circuitry **1300** of FIG. **13**, includes example input/output (I/O) circuitry **1302** to obtain and/or output data to/from example configuration circuitry **1304** and/or external hardware **1306**. For example, the configuration circuitry **1304** may be implemented by interface circuitry that may obtain a binary file, which may be implemented by a bit stream, data, and/or machine-readable instructions, to configure the FPGA circuitry **1300**, or portion(s) thereof. In some such examples, the configuration circuitry **1304** may obtain the binary file from a user, a machine (e.g., hardware circuitry (e.g., programmable or dedicated circuitry) that may implement an Artificial Intelligence/Machine Learning (AI/ML) model to generate the binary file), etc., and/or any combination(s) thereof). In some examples, the external hardware **1306** may be implemented by external hardware circuitry. For example, the external hardware **1306** may be implemented by the microprocessor **1200** of FIG. **12**.

[0098] The FPGA circuitry **1300** also includes an array of example logic gate circuitry **1308**, a plurality of example configurable interconnections **1310**, and example storage circuitry **1312**. The logic gate circuitry **1308** and the configurable interconnections **1310** are configurable to instantiate one or more operations/functions that may correspond to at least some of the machine-readable instructions of FIG. **7** and/or other desired operations. The logic gate circuitry **1308** shown in FIG. **13** is fabricated in blocks or groups. Each block includes semiconductor-based electrical structures that may be configured into logic circuits. In some examples, the electrical structures include logic gates (e.g., And gates, Or gates, Nor gates, etc.) that provide basic building blocks for logic circuits. Electrically controllable switches (e.g., transistors) are present within each of the logic gate circuitry **1308** to enable configuration of the electrical structures and/or the logic gates to form circuits to perform desired operations/functions. The logic gate circuitry **1308** may include other electrical structures such as look-up tables (LUTs), registers (e.g., flip-flops or latches), multiplexers, etc.

[0099] The configurable interconnections **1310** of the illustrated example are conductive pathways, traces, vias, or the like that may include electrically controllable switches (e.g., transistors) whose state can be changed by programming (e.g., using an HDL instruction language) to activate or deactivate one or more connections between one or more of the logic gate circuitry **1308** to program desired logic circuits.

[0100] The storage circuitry **1312** of the illustrated example is structured to store result(s) of the one or more of the operations performed by corresponding logic gates. The storage circuitry **1312** may be implemented by registers or the like. In the illustrated example, the storage circuitry **1312** is distributed amongst the logic gate circuitry **1308** to facilitate access and increase execution speed.

[0101] The example FPGA circuitry **1300** of FIG. **13** also includes example dedicated operations circuitry **1314**. In this example, the dedicated operations circuitry **1314** includes special purpose circuitry **1316** that may be invoked to implement commonly used functions to avoid the need to program those functions in the field. Examples of such special purpose circuitry **1316** include memory (e.g., DRAM) controller circuitry, PCIe controller circuitry, clock circuitry, transceiver circuitry, memory, and multiplier-accumulator circuitry. Other types of special purpose circuitry may be present. In some examples, the FPGA circuitry **1300** may also include example general purpose programmable circuitry **1318** such as an example CPU **1320** and/or an example DSP **1322**.

Other general purpose programmable circuitry **1318** may additionally or alternatively be present such as a GPU, an XPU, etc., that can be programmed to perform other operations.

[0102] Although FIGS. **12** and **13** illustrate two example implementations of the programmable circuitry **1112** of FIG. **11**, many other approaches are contemplated. For example, FPGA circuitry may include an on-board CPU, such as one or more of the example CPU **1320** of FIG. **12**. Therefore, the programmable circuitry **1112** of FIG. **11** may additionally be implemented by combining at least the example microprocessor **1200** of FIG. **12** and the example FPGA circuitry **1300** of FIG. **13**. In some such hybrid examples, one or more cores **1202** of FIG. **12** may execute a first portion of the machine-readable instructions represented by the flowchart of FIG. **7** to perform first operation(s)/function(s), the FPGA circuitry **1300** of FIG. **13** may be configured and/or structured to perform second operation(s)/function(s) corresponding to a second portion of the machine-readable instructions represented by the flowcharts of FIG. **7**, and/or an ASIC may be configured and/or structured to perform third operation(s)/function(s) corresponding to a third portion of the machine-readable instructions represented by the flowcharts of FIG. **7**.

[0103] It should be understood that some or all of the circuitry of FIG. **2** may, thus, be instantiated at the same or different times. For example, same and/or different portion(s) of the microprocessor **1200** of FIG. **12** may be programmed to execute portion(s) of machine-readable instructions at the same and/or different times. In some examples, same and/or different portion(s) of the FPGA circuitry **1300** of FIG. **13** may be configured and/or structured to perform operations/functions corresponding to portion(s) of machine-readable instructions at the same and/or different times.

[0104] In some examples, some or all of the circuitry of FIG. **2** may be instantiated, for example, in one or more threads executing concurrently and/or in series. For example, the microprocessor **1200** of FIG. **12** may execute machine-readable instructions in one or more threads executing concurrently and/or in series. In some examples, the FPGA circuitry **1300** of FIG. **13** may be configured and/or structured to carry out operations/functions concurrently and/or in series. Moreover, in some examples, some or all of the circuitry of FIG. **2** may be implemented within one or more virtual machines and/or containers executing on the microprocessor **1200** of FIG. **12**.

[0105] In some examples, the programmable circuitry **1112** of FIG. **11** may be in one or more packages. For example, the microprocessor **1200** of FIG. **12** and/or the FPGA circuitry **1300** of FIG. **13** may be in one or more packages. In some examples, an XPU may be implemented by the programmable circuitry **1112** of FIG. **11**, which may be in one or more packages. For example, the XPU may include a CPU (e.g., the microprocessor **1200** of FIG. **12**, the CPU **1320** of FIG. **13**, etc.) in one package, a DSP (e.g., the DSP **1322** of FIG. **13**) in another package, a GPU in yet another package, and an FPGA (e.g., the FPGA circuitry **1300** of FIG. **13**) in still yet another package.

[0106] A block diagram illustrating an example software distribution platform **1405** to distribute software such as the example machine-readable instructions **1132** of FIG. **11** to other hardware devices (e.g., hardware devices owned and/or operated by third parties from the owner and/or operator of the software distribution platform) is illustrated in FIG. **14**. The example software distribution platform **1405** may be implemented by any computer server, data facility, cloud service, etc., capable of storing and transmitting software to other computing devices. The third parties may be customers of the entity owning and/or operating the software distribution platform **1405**. For example, the entity that owns and/or operates the software distribution platform **1405** may be a developer, a seller, and/or a licensor of software such as the example machine-readable instructions **1132** of FIG. **11**. The third parties may be consumers, users, retailers, OEMs, etc., who purchase and/or license the software for use and/or re-sale and/or sub-licensing. In the illustrated example, the software distribution platform **1405** includes one or more servers and one or more storage devices. The storage devices store the machine-readable instructions **1132**, which may correspond to the example machine-readable instructions of FIG. **7**, as described above. The one or more servers of the example software distribution platform **1405** are in communication with an example network **1410**, which may correspond to any one or more of the Internet and/or any of the example networks described above. In some examples, the one or more servers are responsive to requests to transmit the software to a requesting party as part of a commercial transaction. Payment for the delivery, sale, and/or license of the software may be handled by the one or more servers of the software distribution platform and/or by a third party payment entity. The servers enable purchasers and/or licensors to download the machine-readable instructions **1132** from the software distribution platform **1405**. For example, the software, which may correspond to the example machine-readable instructions of FIG. **7**, may be downloaded to the example programmable circuitry platform **1100**, which is to execute the machine-readable instructions **1132** to implement the audio creation circuitry **120**. In some examples, one or more servers of the software distribution platform **1405** periodically offer, transmit, and/or force updates to the software (e.g., the example machine-readable instructions **1132** of FIG. **11**) to ensure improvements, patches, updates, etc., are distributed and applied to the software at the end user devices. Although referred to as software above, the distributed "software" could alternatively be firmware.

[0107] "Including" and "comprising" (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of "include" or "comprise" (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase "at least" is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term "comprising" and "including" are open ended. The term "and/or" when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase "at least one of A and B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase "at least one of A or B" is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0108] As used herein, singular references (e.g., "a", "an", "first", "second", etc.) do not exclude a plurality. The term "a" or "an" object, as used herein, refers to one or more of that object. The terms "a" (or "an"), "one or more", and "at least one" are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0109] As used herein, unless otherwise stated, the term "above" describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is "below" a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0110] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0111] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0112] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0113] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of +/−10% unless otherwise specified herein.

[0114] As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0115] As used herein, “programmable circuitry” is defined to include (i) one or more special purpose electrical circuits (e.g., an application specific circuit (ASIC)) structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmable with instructions to perform specific functions(s) and/or operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of programmable circuitry include programmable microprocessors such as Central Processor Units (CPUs) that may execute first instructions to perform one or more operations and/or functions, Field Programmable Gate Arrays (FPGAs) that may be programmed with second instructions to cause configuration and/or structuring of the FPGAs to instantiate one or more operations and/or functions corresponding to the first instructions, Graphics Processor Units (GPUs) that may execute first instructions to perform one or more operations and/or functions, Digital Signal Processors (DSPs) that may execute first instructions to perform one or more operations and/or functions, XPUs, Network Processing Units (NPUs) one or more microcontrollers that may execute first instructions to perform one or more operations and/or functions and/or integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of programmable circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more NPUs, one or more DSPs, etc., and/or any combination(s) thereof), and orchestration technology (e.g., application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of programmable circuitry is/are suited and available to perform the computing task(s).

[0116] As used herein integrated circuit/circuitry is defined as one or more semiconductor packages containing one or more circuit elements such as transistors, capacitors, inductors, resistors, current paths, diodes, etc. For example an integrated circuit may be implemented as one or more of an ASIC, an FPGA, a chip, a microchip, programmable circuitry, a semiconductor substrate coupling multiple circuit elements, a system on chip (SoC), etc.

[0117] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that analyze source media to identify key features of the source media and automatically generate audio tracks that align with the identified features. Example systems, apparatus, articles of manufacture, and methods disclosed herein generate copyright-free music tailored to source media, enabling content creators to efficiently create media without risk of violating copyright rules enforced by various media publishing platforms. Example systems, apparatus, articles of manufacture, and methods disclosed herein enable varying degrees of content creator control over audio track generation to generate audio tracks adapted to the source media without requiring extensive user effort. Disclosed systems, apparatus, articles of manufacture, and methods improve the efficiency of using a computing device by providing a text management structure implementing a prompt generation format for streamlined prompt generation based on video analysis results and user preferences, reducing computing costs (e.g., time, computing resources required, etc.) associated with prompt generation. Furthermore, disclosed systems, apparatus, articles of manufacture, and methods maximize compute resource efficiency of a computing device by detecting the configuration of the system and coordinating workloads based on the configuration, as well as performing workloads at least partially in parallel to reduce total compute time. Disclosed systems, apparatus, articles of manufacture, and methods are accordingly directed to one or more improvement(s) in the operation of a machine such as a computer or other electronic and/or mechanical device.

[0118] Systems, apparatus, articles of manufacture, and methods are disclosed for dynamic music creation in video content creation applications. Further examples and combinations thereof include the following:

[0119] Example 1 includes an apparatus to generate audio tracks for a video content creation application including interface circuitry, machine-readable instructions, and at least one processor circuit to be programmed by the machine-readable instructions to: analyze a video to determine video characteristics; generate a prompt for a machine learning model based on the video characteristics and at least one user preference; provide the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and provide the audio track to a video content creation application.

[0120] Example 2 includes the apparatus of example 1, wherein one or more of the at least one processor circuit is to cause presentation of a user interface, the user interface to allow a user to set the at least one user preference for audio generation.

[0121] Example 3 includes the apparatus of any of examples 1 or 2, wherein one or more of the at least one processor circuit is to: generate, in response to the user updating the at least one user preference to create at least one updated user preference, a second prompt for the machine learning model based on the at least one updated user preference; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and provide the second audio track to the video content creation application.

[0122] Example 4 includes the apparatus of any of examples 1-3, wherein the at least one processor circuit includes a neural processing unit (NPU), a central processing unit (CPU), and a graphics processing unit (GPU).

[0123] Example 5 includes the apparatus of any of examples 1-4, wherein the at least one user preference includes at least one of a genre preference, an audio track duration, a priority of the audio track, and weights for respective video characteristics for prompt generation.

[0124] Example 6 includes the apparatus of any of examples 1-5, wherein the video characteristics include at least one of a presence of speech, a color tone of the video, a type of activity shown, individuals shown, a level of happiness, a type of the video, a speed of action, an environment of the video, and a number of people in the video.
[0125] Example 7 includes the apparatus of any of examples 1-6,
[0126] wherein the machine learning model is a first machine learning model and one or more of the at least one processor circuit is to execute a second machine learning model to analyze the video to determine video characteristics.
[0127] Example 8 includes the apparatus of any of examples 1-7,
[0128] wherein the video is a first video, the video characteristics are first video characteristics, the prompt is a first prompt, the audio track is a first audio track, and one or more of the at least one processor circuit is to: analyze a second video, at least partially in parallel with the first video, to determine second video characteristics; generate a second prompt for the machine learning model based on the second video characteristics; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt, at least partially in parallel with the generation of the first audio track; and provide the second audio track, with the first audio track, to the video content creation application.
[0129] Example 9 includes at least one non-transitory machine-readable medium including machine-readable instructions to cause at least one processor circuit to at least: analyze a video to determine video characteristics; generate a prompt for a machine learning model based on the video characteristics and at least one user preference parameter; provide the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and provide the audio track to a media creation application.
[0130] Example 10 includes the storage medium of example 9, wherein the machine-readable instructions are to cause one or more of the at least one processor circuit to at least cause presentation of a user interface, the user interface to allow a user to set the at least one user preference parameter for audio generation.
[0131] Example 11 includes the storage medium of any of examples 9 or 10, wherein the machine-readable instructions are to cause one or more of the at least one processor circuit to: generate, in response to the user updating the at least one user preference parameter to create at least one updated user preference parameter, a second prompt for the machine learning model based on the at least one updated user preference parameter; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and provide the second audio track to the media creation application.
[0132] Example 12 includes the storage medium of any of examples 9-11, wherein the at least one processor circuit includes a neural processing unit (NPU), a central processing unit (CPU), and a graphics processing unit (GPU).
[0133] Example 13 includes the storage medium of any of examples 9-12, wherein the at least one user preference parameter includes at least one of a genre preference, an audio track duration, a priority of the audio track, and weights for respective video characteristics for prompt generation.
[0134] Example 14 includes the storage medium of any of examples 9-13, wherein the video characteristics include at least one of a presence of speech, a color tone of the video, a type of activity shown, individuals shown, a level of happiness, a type of the video, a speed of action, an environment of the video, and a number of people in the video.
[0135] Example 15 includes the storage medium of any of examples 9-14, wherein the machine learning model is a first machine learning model and the machine-readable instructions are to cause one or more of the at least one processor circuit to execute a second machine learning model to analyze the video to determine video characteristics.
[0136] Example 16 includes the storage medium of any of examples 9-15, wherein the video is a first video, the video characteristics are first video characteristics, the prompt is a first prompt, the

audio track is a first audio track, and the machine-readable instructions are to cause one or more of the at least one processor circuit to: analyze a second video, at least partially in parallel with the first video, to determine second video characteristics; generate a second prompt for the machine learning model based on the second video characteristics; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt, at least partially in parallel with the generation of the first audio track; and provide the second audio track, with the first audio track, to the media creation application.

[0137] Example 17 includes a method for audio track generation for video content, including: analyzing a video to determine video characteristics; generating, by at least one processor circuit programmed by at least one instruction, a prompt for a machine learning model based on the video characteristics and at least one user preference parameter; providing, by one or more of the at least one processor circuit, the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and providing the audio track to a media creation application.

[0138] Example 18 includes the method of example 17, further including causing presentation of a user interface, the user interface to allow a user to set the at least one user preference parameter for audio generation.

[0139] Example 19 includes the method of any of examples 17 or 18, further including: generating, in response to the user updating the at least one user preference parameter to create at least one updated user preference parameter, a second prompt for the machine learning model based on the at least one updated user preference parameter; providing the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and providing the second audio track to the media creation application.

[0140] Example 20 includes the method of any of examples 17-19, wherein analyzing the video to determine video characteristics includes executing a second machine learning model.

[0141] Example 21 includes the method of any of examples 17-20, wherein the at least one processor circuit includes a neural processing unit (NPU), a central processing unit (CPU), and a graphics processing unit (GPU).

[0142] Example 22 includes the method of any of examples 17-21, wherein the at least one user preference parameter includes at least one of a genre preference, an audio track duration, a priority of the audio track, and a weight for the respective video characteristics for prompt generation.

[0143] Example 23 includes the method of any of examples 17-22, wherein the video characteristics include at least one of a presence of speech, a color tone of the video, a type of activity shown, individuals shown, a level of happiness, a type of the video, a speed of action, an environment of the video, and a number of people in the video.

[0144] Example 24 includes the method of any of examples 17-23, wherein the video is a first video, the video characteristics are first video characteristics, the prompt is a first prompt, the audio track is a first audio track, further including: analyzing a second video in parallel with the first video to determine second video characteristics; generating a second prompt for the machine learning model based on the second video characteristics; providing the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt, in parallel with the generation of the first audio track; and providing the second audio track, with the first audio track, to the media creation application.

[0145] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

1. An apparatus to generate audio tracks for a video content creation application, comprising: interface circuitry; machine-readable instructions; and at least one processor circuit to be programmed by the machine-readable instructions to: analyze a video to determine video characteristics; generate a prompt for a machine learning model based on the video characteristics and at least one user preference; provide the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and provide the audio track to a video content creation application.
2. The apparatus of claim 1, wherein one or more of the at least one processor circuit is to cause presentation of a user interface, the user interface to allow a user to set the at least one user preference for audio generation.
3. The apparatus of claim 2, wherein one or more of the at least one processor circuit is to: generate, in response to the user updating the at least one user preference to create at least one updated user preference, a second prompt for the machine learning model based on the at least one updated user preference; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and provide the second audio track to the video content creation application.
4. The apparatus of claim 1, wherein the at least one processor circuit includes a neural processing unit (NPU), a central processing unit (CPU), and a graphics processing unit (GPU).
5. The apparatus of claim 1, wherein the at least one user preference includes at least one of a genre preference, an audio track duration, a priority of the audio track, and weights for respective video characteristics for prompt generation.
6. The apparatus of claim 1, wherein the video characteristics include at least one of a presence of speech, a color tone of the video, a type of activity shown, individuals shown, a level of happiness, a type of the video, a speed of action, an environment of the video, and a number of people in the video.
7. The apparatus of claim 1, wherein the machine learning model is a first machine learning model and one or more of the at least one processor circuit is to execute a second machine learning model to analyze the video to determine video characteristics.
8. The apparatus of claim 1, wherein the video is a first video, the video characteristics are first video characteristics, the prompt is a first prompt, the audio track is a first audio track, and one or more of the at least one processor circuit is to: analyze a second video, at least partially in parallel with the first video, to determine second video characteristics; generate a second prompt for the machine learning model based on the second video characteristics; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt, at least partially in parallel with the generation of the first audio track; and provide the second audio track, with the first audio track, to the video content creation application.
9. At least one non-transitory machine-readable medium comprising machine-readable instructions to cause at least one processor circuit to at least: analyze a video to determine video characteristics; generate a prompt for a machine learning model based on the video characteristics and at least one user preference parameter; provide the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and provide the audio track to a media creation application.
10. The at least one non-transitory machine-readable medium of claim 9, wherein the machine-readable instructions are to cause one or more of the at least one processor circuit to at least cause presentation of a user interface, the user interface to allow a user to set the at least one user preference parameter for audio generation.
11. The at least one non-transitory machine-readable medium of claim 10, wherein the machine-

readable instructions are to cause one or more of the at least one processor circuit to: generate, in response to the user updating the at least one user preference parameter to create at least one updated user preference parameter, a second prompt for the machine learning model based on the at least one updated user preference parameter; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and provide the second audio track to the media creation application.

12. The at least one non-transitory machine-readable medium of claim 9, wherein the at least one processor circuit includes a neural processing unit (NPU), a central processing unit (CPU), and a graphics processing unit (GPU).

13. The at least one non-transitory machine-readable medium of claim 9, wherein the at least one user preference parameter includes at least one of a genre preference, an audio track duration, a priority of the audio track, and weights for the respective video characteristics for prompt generation.

14. The at least one non-transitory machine-readable medium of claim 9, wherein the video characteristics include at least one of a presence of speech, a color tone of the video, a type of activity shown, individuals shown, a level of happiness, a type of the video, a speed of action, an environment of the video, and a number of people in the video.

15. The at least one non-transitory machine-readable medium of claim 9, wherein the machine learning model is a first machine learning model and the machine-readable instructions are to cause one or more of the at least one processor circuit to execute a second machine learning model to analyze the video to determine video characteristics.

16. The at least one non-transitory machine-readable medium of claim 9, wherein the video is a first video, the video characteristics are first video characteristics, the prompt is a first prompt, the audio track is a first audio track, and the machine-readable instructions are to cause one or more of the at least one processor circuit to: analyze a second video, at least partially in parallel with the first video, to determine second video characteristics; generate a second prompt for the machine learning model based on the second video characteristics; provide the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt, at least partially in parallel with the generation of the first audio track; and provide the second audio track, with the first audio track, to the media creation application.

17. A method for audio track generation for video content, comprising: analyzing a video to determine video characteristics; generating, by at least one processor circuit programmed by at least one instruction, a prompt for a machine learning model based on the video characteristics and at least one user preference parameter; providing, by one or more of the at least one processor circuit, the prompt to machine learning model execution circuitry to cause generation of an audio track based on the prompt; and providing the audio track to a media creation application.

18. The method of claim 17, further including causing presentation of a user interface, the user interface to allow a user to set the at least one user preference parameter for audio generation.

19. The method of claim 18, further including: generating, in response to the user updating the at least one user preference parameter to create at least one updated user preference parameter, a second prompt for the machine learning model based on the at least one updated user preference parameter; providing the second prompt to the machine learning model execution circuitry to cause generation of a second audio track based on the second prompt; and providing the second audio track to the media creation application.

20. The method of claim 17, wherein analyzing the video to determine video characteristics includes executing a second machine learning model.